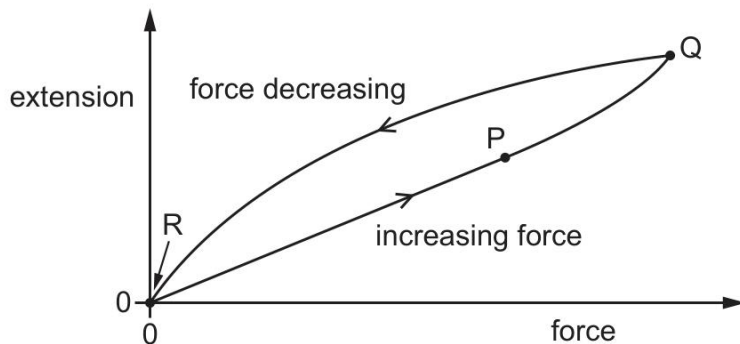


22 The graph shows the variation with force of the extension of a wire.



The force is gradually increased to a maximum at Q and then gradually decreased to zero at R.

Which statement is correct?

- A** Along the line PQ, the wire obeys Hooke's law.
- B** Along the line RP, the spring constant is equal to $\frac{\text{extension}}{\text{force}}$.
- C** The wire has elastic deformation at point Q.
- D** The work done in stretching the wire to P is equal to (force $\times$ extension) at point P.